Studio

Plumbing

Everything the plumber needs for one container studio: supply, hot water, drains, the tests, and what the studio needs from the farm.

Status: revision A2, 2026-10-28, PRELIMINARY. No guest stays before the septic system passes its inspection (D-016).

The two studios are piped the same, so one page serves both.

Kinyarwanda: soma aya mabwiriza mu Kinyarwanda · Print: this page as a PDF · The pipes drawn: the studio page · When it happens: the execution plan

At a glance

- **End A**: the closed end wall, by the bed. **End B**: the cargo-door end, by the bathroom.
- **Back wall**: the long wall with no openings. **Front wall**: the long wall with the sliding glass door.
- Positions in the fittings table are millimetres from the finished end-A wall (x), from the finished back wall (y) and up from the finished floor (z).

What	Design
Supply pipe	PPR PN20, 25 mm for the main, 20 mm inside
Main in	under the floor at end B, up into the sink base: stop valve, Y-filter, check meter (SV)
Hot water	an instant heater, 5.5 kW, IPX4, on end wall B inside the shower, beside the hand shower; no tank. It heats about 4 L/min to 40 °C, set on its own display, for the shower only. The sink and the basin are cold (D-053)
Drains	WC 110 mm, shower 50 mm, basin 40 mm, kitchen sink 50 mm on its own line to a grease trap; all at 1:40 or steeper
Vent	75 mm open vent up end B, ending 900 mm above the roof guardrail
Shower tray	1000 × 900
Pressure	2.5 to 3 bar from the farm's booster pump

When	What	Step
After the battens (S03)	First fix: main in, every supply and drain run, the 10 bar test	S04
With the bathroom	Tray (S09), WC, heater and basin (S10)	S09, S10
With the kitchen	Sink, tap and filter	S13

From the farm

- One cold supply at 2.5 to 3 bar from the farm's storage tank and booster pump. Mains water enters that tank through an air gap, so private water never flows back into the main (D-016).
- A soil connection from the inspection chamber outside end B to the farm's septic tank, and the kitchen line to a grease trap with its own soak pit.
- The septic tank sits at least 50 m from the borehole, 3 to 7 m from buildings and the boundary, downhill of the borehole, within 30 m of tanker access, and only after a percolation test.

Site water and septic sizing for the two studios · download CSV

item	value
persons	4
daily_L	600
tank_m3	2.5
septic_rs143_m3	3
septic_2012_m3	2.72
septic_uk_m3	3.08
septic_m3	3.5
chamber1_m3	2.33
chamber2_m3	1.17
pump_lpm	26

Hold points

1. **Battens first.** Start the first fix only when the battens (S03) are fixed and the floor sheets are down.
2. **The 10 bar test.** Fill the PPR, hold 10 bar for one hour with no drop, write it down and sign it before the spray foam (S05). The foam buries the pipes.
3. **Before guests.** The drains are fill-tested, the vent smoke-tested, the shower measured with a thermometer, and the farm's septic system inspected.

Steps

S04: first fix

1. Bring the cold main (W1, 25 mm) in under the floor at end B and up into the sink base through the stop valve, Y-filter and check meter, where it can be shut off without tools.
2. Run the cold inside the units and the walls: through the partition, the back wall's lining and end wall B to the WC cistern (W4), the basin (W5) and up to the heater's foot in the shower (W2). Keep cold 300 mm above the floor.
3. No hot pipes: the only hot water is the hand shower's hose from the top of the heater (H1).
4. Every run goes straight: along the wall, then straight up to its outlet. No diagonal legs.
5. Hang the drains in the void under the floor (the container stands about 700 mm up): WC 110 mm, shower 50 mm, basin 40 mm to the inspection chamber; the kitchen sink 50 mm on its own line to the grease trap. Pass between the 122 mm cross-members, never through them.
6. Fall at 1:40 or steeper; rodding eyes at every bend; the 75 mm vent up end B.
7. Pressure-test at 10 bar for one hour, photograph every wall, sign the release for the foam.

S09, S10 and S13: second fix

1. S09: set the shower tray (1000 × 900) on its 90 mm waste, with the carpenter's ledge and the fixed glass screen.
2. S10: hang the wall-hung WC on its slim concealed cistern (80 mm frame in end wall B, 30 mm build-out, dual-flush plate).
3. S10: hang the 5.5 kW instant heater (200 × 330 × 90) on end wall B inside the shower, between the slide rail and the glass, its display about 1.3 m up; connect the cold at its foot and the hand shower's hose at its top. It has no tank, so it needs no relief valve.
4. S10: fit the basin (400 × 300) and its cold tap with a bottle trap in the wash lobby; the slide rail and the hand shower on end wall B.

5. S13: the kitchen tap (cold) on its angle valve, and the drinking-water filter in the sink base with its own faucet by the tap.

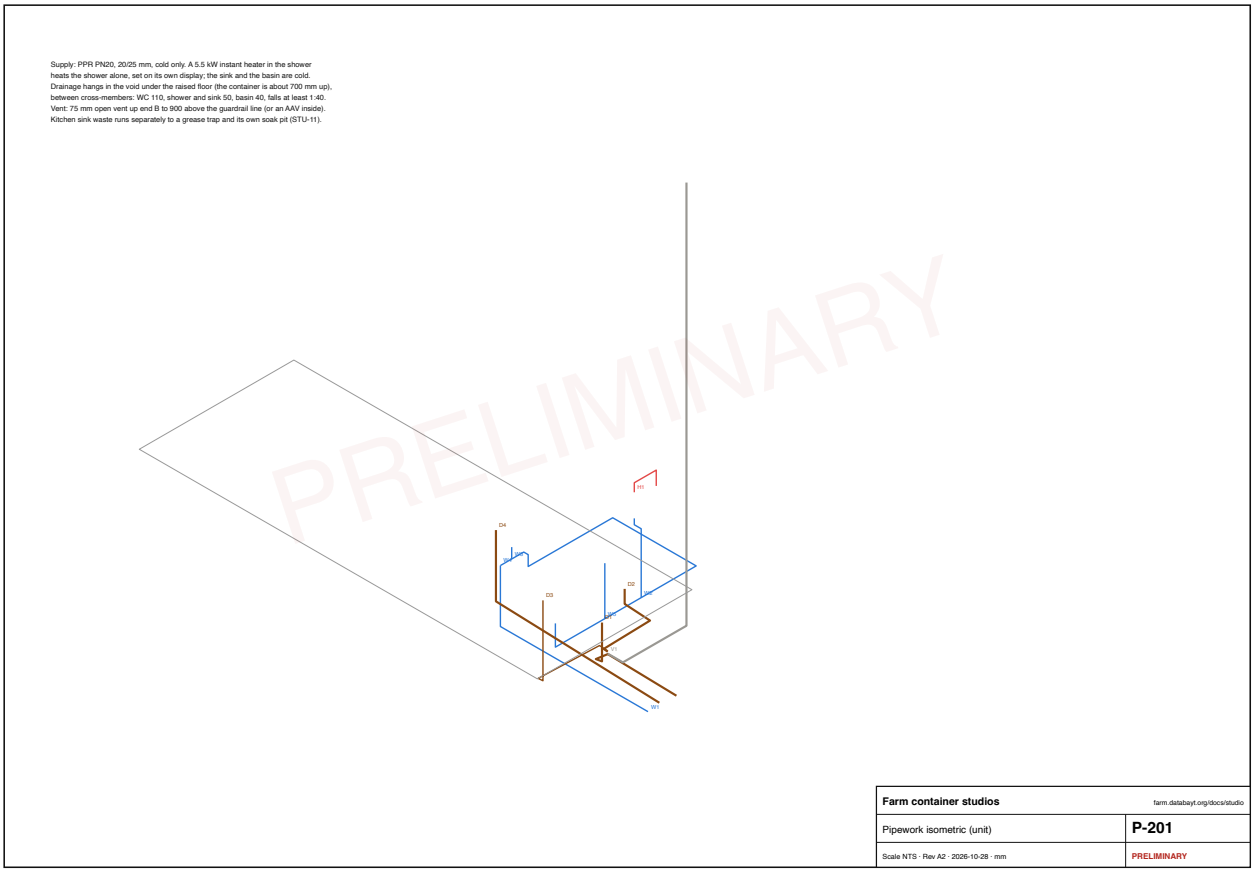
Tests

1. PPR at 10 bar for one hour, before the foam (S04).
2. Fill-test every drain and check every trap holds its seal.
3. Smoke-test the vent.
4. Run the shower at full flow and measure it with a thermometer: about 40 °C. If it is hotter, turn the heater's display down.
5. Run a long shower: the water stays hot, since the heater has no tank to empty.

Runs

Each run is checked by the model: every leg runs along a wall or straight up or down, each drain falls at least 1:40.

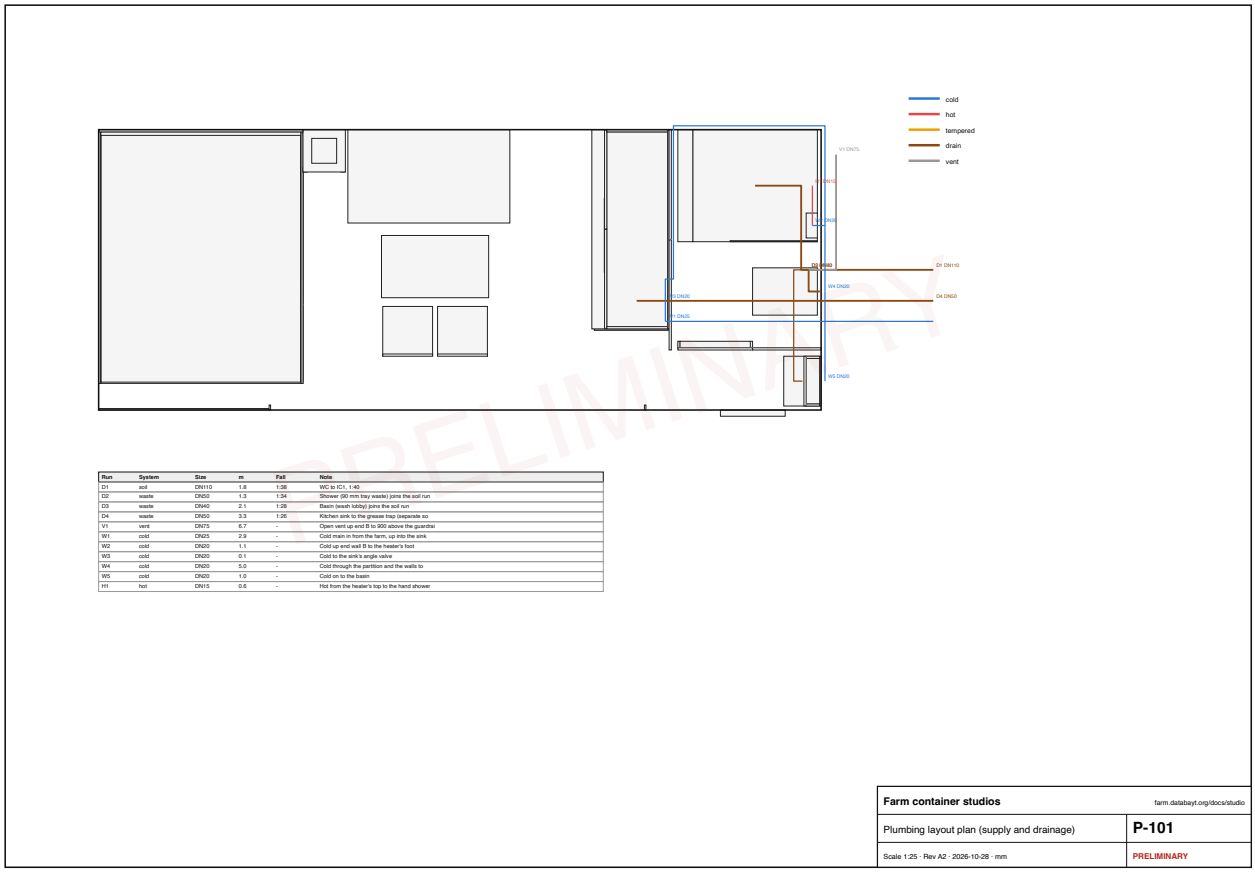
Run	System	Size	Length	Fall	What it does
D1	soil	DN110	1.8 m	1:38	WC to IC1, 1:40
D2	waste	DN50	1.3 m	1:34	Shower (90 mm tray waste) joins the soil run
D3	waste	DN40	2.1 m	1:28	Basin (wash lobby) joins the soil run
D4	waste	DN50	3.3 m	1:26	Kitchen sink to the grease trap (separate soak pit)
V1	vent	DN75	6.7 m	–	Open vent up end B to 900 above the guardrail
W1	cold	DN25	2.9 m	–	Cold main in from the farm, up into the sink base: stop valve, filter, check meter
W2	cold	DN20	1.1 m	–	Cold up end wall B to the heater's foot
W3	cold	DN20	0.1 m	–	Cold to the sink's angle valve
W4	cold	DN20	5 m	–	Cold through the partition and the walls to the WC cistern
W5	cold	DN20	1 m	–	Cold on to the basin
H1	hot	DN15	0.6 m	–	Hot from the heater's top to the hand shower: its own hose



P-201 · Pipework isometric (unit) · open full size

Points

10 points, from the stop valve on the main to the vent above the roof. The sink, the heater, the shower, the WC and the basin sit within about 1.5 m of each other, so the pipes stay short.



Run	System	Size	m	Fall	Note
D1	soil	DN110	1.8	1:38	WC to IC1, 1:40
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W2	cold	DN20	1.1	-	Cold up and wall B to the heater's foot
W3	cold	DN20	0.1	-	Cold to the sink's angle valve
W4	cold	DN20	5.0	-	Cold through the partition and the walls to
W5	cold	DN20	1.0	-	Cold on to the basin
H1	hot	DN15	0.6	-	Hot from the heater's top to the hand shower

P-101 · Plumbing layout plan (supply and drainage) · open full size

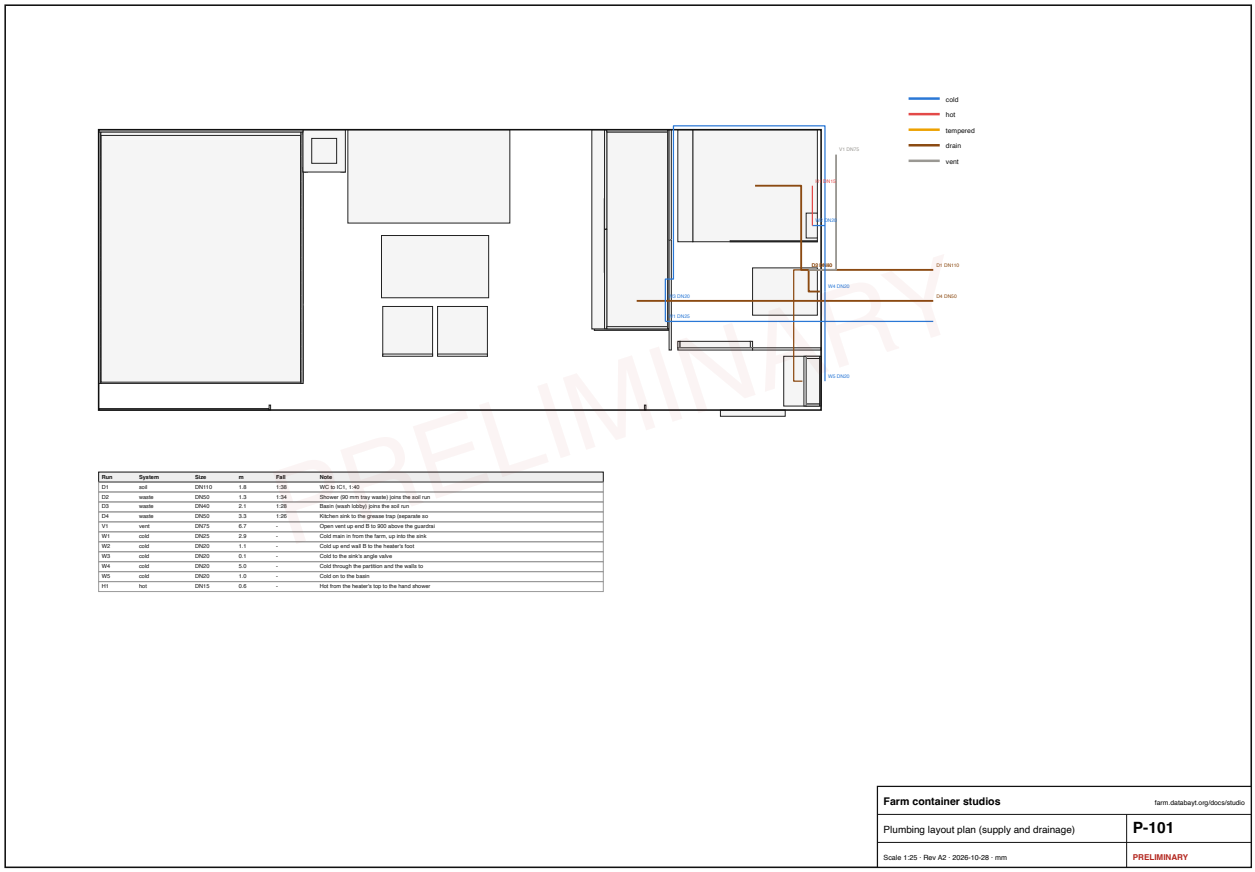
Point	What	Type	Pipes	Height	Rating
SV	Stop valve, Y-filter and check meter on the main, in the sink base	valve	W1	320 mm	DN25
WH	Instant heater on end wall B inside the shower, its display at eye height; hot only to the shower	heater	W2 H1	1,415 mm	5.5 kW, about 4 L/min, 40 °C
KT	Kitchen tap (cold), its angle valve under the sink	tap	W3 D4	900 mm	6 L/min
DF	Drinking-water filter in the sink base, its own faucet by the tap	filter	W3	350 mm	drinking water

Point	What	Type	Pipes	Height	Rating
SH	Hand shower on a slide rail on end wall B, fed by the heater's hose	shower	H1 D2	1,110 mm	about 4 L/min
WC	Concealed cistern in end wall B, dual-flush plate	WC	W4 D1	1,010 mm	dual flush
BT	Basin tap (cold), its angle valve under the console	tap	W5 D3	820 mm	6 L/min
IC1	Inspection chamber outside end B, on to the farm's septic tank	on the site	D1 D2 D3	−326 mm	DN110 in
GT	To the farm's grease trap and its own soak pit	on the site	D4	−290 mm	DN50 in
VT	Vent terminal, 900 mm above the roof guardrail	vent	V1	5,300 mm	DN75

Open questions

1. Where does the farm's septic tank go, and has its percolation test passed? The studios' drains and their first guests wait on both (**the farm's water**).

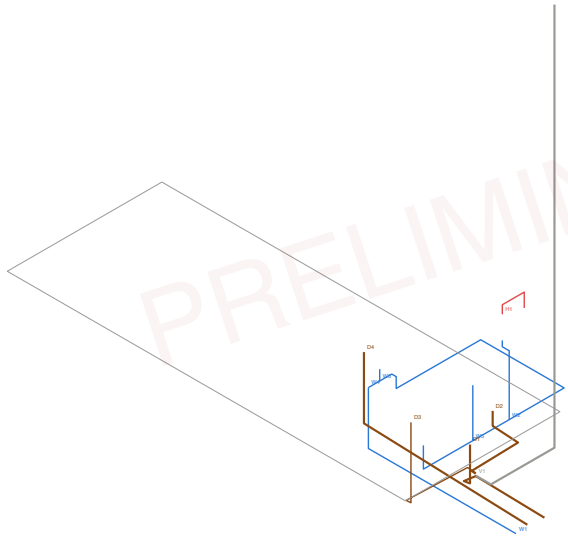
Drawings



Run	System	Size	m	Fall	Note
D1	soil	DN110	1.8	1:38	WC to IC1, 1:40
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W4	cold	DN20	5.0	-	Cold through the partition and the walls to
W5	cold	DN20	1.0	-	Cold on to the basin
H1	hot	DN15	0.6	-	Hot from the heater's top to the hand shower

P-101 Plumbing layout plan · open full size

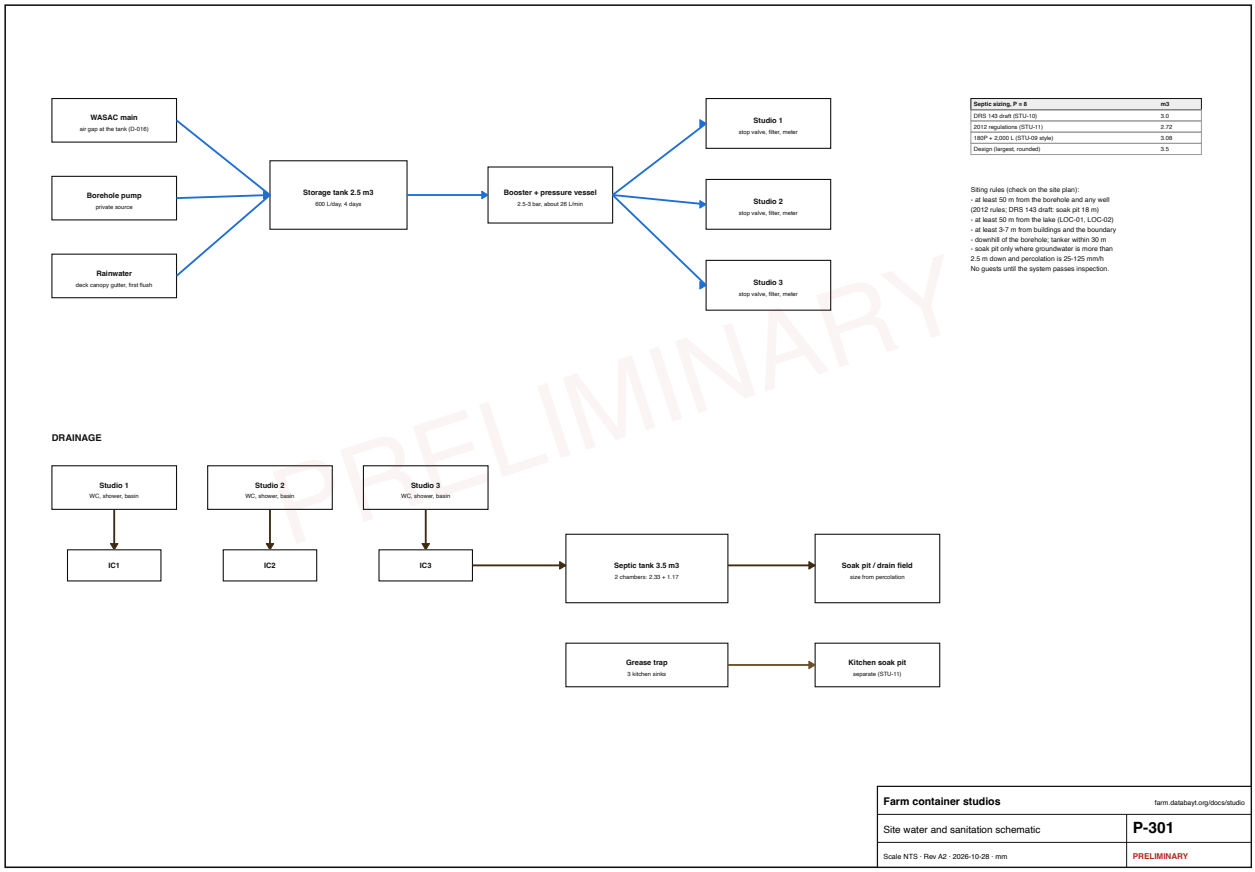
Supply: PPR PN20, 20/25 mm, cold only. A 5.5 kW instant heater in the shower heats the shower alone, set on its own display; the sink and the basin are cold.
Drainage hangs in the void under the raised floor (the container is about 700 mm up), between cross-members: WC 110, shower and sink 50, basin 40, falls at least 1:40.
Vent: 75 mm open vent up end B to 900 above the guardrail line (or an AAV inside).
Kitchen sink waste runs separately to a grease trap and its own soak pit (STU-11).



PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Pipework isometric (unit)	**P-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

P-201 Pipework isometric · open full size



P-301 Site water and sanitation schematic · open full size